

Term Project

Project Due on Jun 28th, 2026

Description:

To Design and Implement a working prototype of a Decision Support System (DSS). This decision support system needs to be 'realistic' and the software should be robust. It can be developed using any language or use any tool. It could also integrate and use multiple languages/packages.

You are required to provide the instructor with a one-page outline of your system **by May 30th, 11:59 PM EST, 2026**. Please discuss your project with the instructor if you have any questions.

Your project report along with the DSS Tool is due on **June 28th, 11:59 PM EST, 2026**. Each team is required to submit executable files or files, a detailed report that addresses domain and problem descriptions, the project's objective, systems design, process flow, etc. Please use figures and flowcharts when needed. The inputs and outputs to the system must be described along with (at least) one case study. The report should not exceed 10 pages (double spaced – 12 point Arial font – one inch margins).

Cruise Line Decision Support System (DSS)

Instructions:

- The students should also make any assumptions necessary to flush out the details.
- It is suggested the student team visit some websites of cruise companies to understand what is required in order to do Cruise DSS.

***This is a fictional case study, any similarity to any real enterprise out there is purely coincidental. ***

Introduction:

ABC Cruise Lines operates cruises out of New York, NY to Canada as well as 'cruises to somewhere', which are 1 day or ½ day cruises to international waters so that patrons can gamble. The Canadian cruises are 7-night or 9-night cruises. The company owns three ships. ABC Cruise Lines (ABC) is headquartered in Binghamton, NY and has approximately 100 shore-side employees as well as 300 ship-board employees. Currently, all reservations are handled through third-parties for which ABC pays commissions. The Enterprise wants to develop their own in-house reservation capabilities and work directly with potential customers and a Decision Support System (DSS) for the operational managers to make decisions.

To understand the project goals, the systems analyst Doug, has an interview with Jane Smith. Jane is the Director of Operations who is sponsoring this project.

Interview:

Doug	Hi, I'm Doug, thanks for taking the time to sit down with me and explain the goals of the project.
Jane	Yes, well I'm happy to finally get started on this project. The DSS will help us manage our costs, project revenue. This can greatly reduce our costs because we currently pay the travel agents a commission of 10%, and up to

15% for some cruises. Currently, 100% of our reservations are done this way, which means we pay out approximately \$10 million in commissions each year. We benchmarked other cruise lines, and industry wide, the larger cruise lines capture about 50% of the bookings themselves for which they pay no commission. If we match the industry average, then we would save \$4 million each year. This is a significant cost savings. An inhouse reservation systems will also enable the DSS to be effective, which will help us better manage operations.

Doug Yes, I would agree. Are there any drawbacks or hurdles to setting up DSS in-house?

Jane Well, to begin with for the reservation system to be brought in house, the travel agents don't like it - but this is the industry trend and there's not much they can do about it. In the airline industry, most airlines don't even pay commissions anymore, so the travel agents charge the customers a service fee.

Doug Why haven't you developed DSS capabilities sooner?

Jane We just outsourced it so that we could concentrate on running the ships. Now we believe we can take it in-house to reduce costs.

Doug Approximately how much do you expect it to cost to develop a DSS and then to operate it each year?

Jane We did some preliminary analysis and believe we could establish a reservation system and then a DSS for less than \$2 million. For operations, we might need about 8-12 reservation agents, a supervisor, 1-2 Analysts and BI Manager and some IT support.

Doug OK, that's good, I'd like to see the analysis but this gives me an ideas of the project budget. How about the project time-frame? What is the goal for having a DSS system in place up and running?

Jane We would hope to deploy the system by May/11/2022. (It is now January so this is a 5-month project)

Doug What exactly must the DSS system do?

Jane In a nutshell, first, the reservation system needs to book reservations. The reservation process involves a lot of steps. For the cruise to somewhere it is relatively simple, but for the 7-night and 9-night cruises the passengers have to select not only the cruise, but the departure date, cabin, transportation if needed, meal plan, onboard credit, and port excursions. Also, we need to be able to handle different types of customers. For example, we get frequent corporate groups for our cruises to somewhere. These groups are handled differently. This should feed a DSS system which will dashboards which will help managers make decisions.

Doug OK, I'm getting this all down. Can you tell me the start and finish of the reservation process as you see it?

Jane It starts when a prospective customer contacts us and it ends once they have booked a cruise.

Doug Does this include paying for the cruise?

Jane Yes, we don't have a billing department that handles that. So, payment is included in the reservation process.

Doug What other systems would reservation have to integrate with?

Jane The revenue management system sets the prices for the cruises. Oh, it also sends information to the revenue management system which might update the pricing depending on how bookings are going. Just prior to the departure of a cruise all the confirmed bookings need to be transferred to the shipboard systems. It also feeds the DSS which helps the operational managers make decisions.

Doug How will we know whether the project is successful?

Jane It is successful if we direct enough booking traffic to it to save money. I think to do this the system will have to be user-friendly and easily accessible. A DSS which can help us to visualize opportunities for improvements. Of course, we'll have to market and promote the service so that people know its available.

Doug I assume a whole new organizational group will be needed?

Jane Yes, part of the project is to design the jobs so that human resources can start recruiting people with the needed skills.

Doug Do you currently have a database for reservations?

Jane No, not really, since it is totally outsourced we never developed one.

Doug OK, I think I have enough information to create a project charter. Once its approved we can get started on this project.

Jane Great, I'm here to help get this started.

Grading Scheme

One page outline submitted by **May 30th, 2026**

Term project as submitted by **Jun 28th, 2026**

Total Value of Term Project for the Course - 20%